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Assignment 4 ME 373

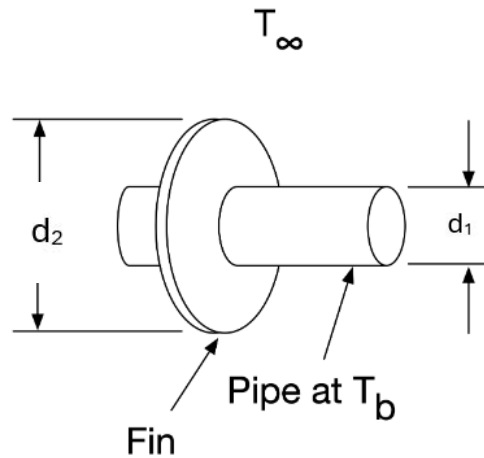
Instructions:

1. Show **all** steps and work in symbolic variables wherever possible. Make sure to clearly state any assumptions/approximations that are not given in the problem description.
2. For problems that ask for plots/graphs, make sure to label all graphs (i.e., the x and y axes both need to be labeled). If you include multiple lines/series on the same graph, clearly indicate what each curve/series corresponds to using a legend or direct labels. Include a caption for each plot, so we know what is being shown.
3. For problems that require code, include the code in your submission by either copy-pasting it in, or printing from your Python editor.
4. Unless you are told to use a built-in integrator like `ode45`, you must implement the methods mentioned yourselves.
5. Submit your assignment as a single combined PDF on Gradescope.
6. These are group assignments of **two** members. More than two members are not allowed. One member must submit the assignment and add the other member as per the instructions, https://www.youtube.com/watch?v=rue7p_kATLA. You can work individually if you prefer.
7. All student names in the group should be clearly listed on the 1st page.
8. The respective solution pages for each question/subquestion must be selected by the student when uploading to Gradescope. If not, 10 points will be reduced.

1. [20 points] The temperature distribution $T(r)$ in an annular fin of inner radius r_1 and outer radius r_2 is described by the equation,

$$r \frac{d^2 T}{dr^2} + \frac{dT}{dr} - rm^2(T - T_\infty) = 0$$

where r is the radial distance from the centerline (the independent variable) and m^2 is a constant that depends on the heat transfer coefficient, thermal conductivity, and thickness of the annulus. Assuming we choose a spatial step size Δr .



- Write the finite-difference representation of the ODE.
- Present the recursion formula of the equation derived in part a.
- The boundary condition at the outer radius $r = r_2$ is described by convection heat transfer:

$$-k \left. \frac{dT}{dr} \right|_{r=r_2} = h[T(r = r_2) - T_\infty]$$

Write the boundary condition at $r = r_2$ in recursion form (i.e., the equation you would implement into your system of equations to solve for temperature).

2. [40 points] Compute and plot the steady-state distribution of concentration for the tank shown in Figure 1. The PDE governing this system is

$$D \left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} \right) - kc = 0 \quad (1)$$

and the boundary conditions are as shown. Employ a value of 0.5 for D and 0.1 for k . Use a small enough $\Delta x = \Delta y = 0.5$ so that your answer does not depend on the value. The wall boundary conditions correspond to zero derivative in the direction normal to the wall.

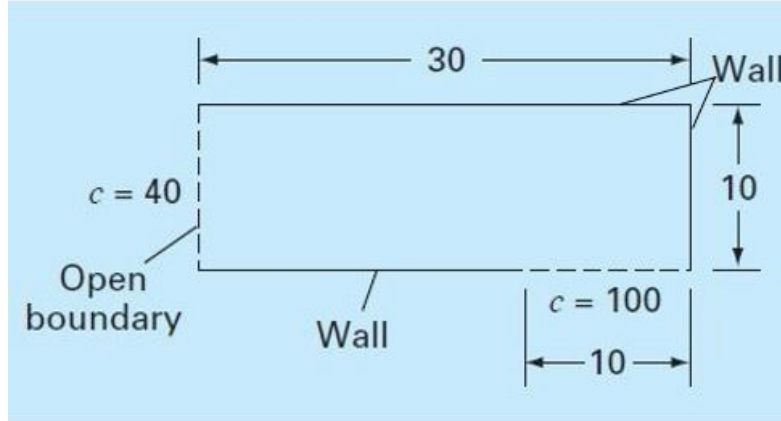


Figure 1: Tank for problem 2

3. [40 points] The advection–diffusion equation is used to compute the distribution of concentration along the length of a rectangular chemical reactor:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - U \frac{\partial c}{\partial x} - kc, \quad (2)$$

where c = concentration (mg/m^3), t = time (min), D = a diffusion coefficient (m^2/min), x = distance along the tank's longitudinal axis (m) where $x = 0$ at the tank's inlet, U = velocity in the x direction (m/min), and k = a reaction rate (min^{-1}) whereby the chemical decays to another form.

Develop an explicit scheme to solve this equation numerically, i.e. central difference in space and forward difference in time. Test it for $k = 0.15$, $D = 100$, and $U = 1$ for a tank of length 10 m. Use a $\Delta x = 0.5$ m, and a step size $\Delta t = 0.001$ min. Assume that the inflow concentration is 100, at the outlet the concentration is not changing with x position, and that the initial concentration in the tank is zero. Perform the simulation from $t = 0$ to 100 min and plot the final resulting concentrations versus x .

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1) considering the governing equation: $r \frac{d^2 T}{dr^2} + \frac{dT}{dr} - r m^2 (T - T_\infty) = 0$

using finite-Difference:

* Deriving *
$$f(x+\Delta x) = f(x) + \Delta x f'(x) + \frac{\Delta x^2}{2!} f''(x) + \dots + \frac{\Delta x^n}{n!} f^n(x)$$

$$f(x-\Delta x) = f(x) - \Delta x f'(x) + \frac{\Delta x^2}{2!} f''(x) - \dots + \frac{\Delta x^n}{n!} f^n(x)$$

$$\therefore f(x+\Delta x) - f(x-\Delta x) = 2\Delta x f'(x) + \frac{2\Delta x^3}{3!} f'''(x)$$

solve for $f'(x)$: $\Rightarrow f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \frac{2\Delta x^3 f'''(x)}{3! 2\Delta x}$

$$\Rightarrow f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \frac{\Delta x^2 f'''(x)}{3!}$$

* Note: $f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$ is the Approximation

+ $\frac{f'''(x)}{3!}$ can be generalized as a small angle Approximation of the remainder.

$$\therefore f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \mathcal{O}(\Delta x^2) \quad \text{second Order accuracy.}$$

Solve for $f''(x)$ such that

$$f(x+\Delta x) + f(x-\Delta x) = 2f(x) + \Delta x^2 f''(x) + \frac{\Delta x^4}{2!} f''''(x)$$

$$\Rightarrow f''(x) = \frac{f(x+\Delta x) + f(x-\Delta x) - 2f(x)}{\Delta x^2} - \frac{\Delta x^4 f''''(x)}{2! \Delta x^2}$$

Similarly: 2nd order Accurate Approximation Small angle / Remainder

∴ Finite Difference general form for 1st and 2nd derivatives

$$f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} \quad + \quad f''(x) = \frac{f(x+\Delta x) + f(x-\Delta x) - 2f(x)}{\Delta x^2}$$

∴
$$r_i \left(\frac{T(r+\Delta r) + T(r-\Delta r) - 2T(r)}{\Delta r^2} \right) + \left(\frac{T(r+\Delta r) - T(r-\Delta r)}{2\Delta r} \right) - r_i m^2 \Delta r^2 (T_i - T_\infty) = 0$$

⇒ finite Difference representation of ODE

*Note: $f(x+\Delta x) \Rightarrow y_{i+1}$, $f(x-\Delta x) \Rightarrow y_{i-1}$, $f(x) \Rightarrow y_i$

∴
$$r_i \frac{T_{i+1} + T_{i-1} - 2T_i}{\Delta r^2} + \frac{T_{i+1} - T_{i-1}}{2\Delta r} - r_i m^2 (T_i - T_\infty) = 0$$

∴ The discretized finite Difference representation of ODE

$$\Rightarrow r_i (T_{i+1} + T_{i-1} - 2T_i) + \frac{\Delta r}{2} (T_{i+1} - T_{i-1}) - r_i m^2 (T_i - T_\infty) \Delta r^2 = 0$$

b) Solving for recursion

$$\Rightarrow r_i T_{i+1} + r_i T_{i-1} - 2r_i T_i + \frac{\Delta r}{2} T_{i+1} - \frac{\Delta r}{2} T_{i-1} - r_i m^2 (T_i - T_\infty) \Delta r^2 = 0$$

general recursion form for finite-Difference

$$T_{i+1} \left(r_i + \frac{\Delta r}{2} \right) - T_i (2r_i + r_i m^2 \Delta r^2) + T_{i-1} \left(r_i - \frac{\Delta r}{2} \right) = T_\infty r_i m^2 \Delta r^2$$

$$\Rightarrow T_{i+1} \left(r_i + \frac{\Delta r}{2} \right) + T_{i-1} \left(r_i - \frac{\Delta r}{2} \right) - T_i (2r_i + r_i m^2 \Delta r^2) + T_\infty r_i m^2 \Delta r^2 = 0$$

$$\therefore T_i = \frac{T_{i+1} \left(r_i + \frac{\Delta r}{2} \right) + T_{i-1} \left(r_i - \frac{\Delta r}{2} \right) + T_\infty r_i m^2 \Delta r^2}{(2r_i + r_i m^2 \Delta r^2)}$$

+

$$\therefore T_{i+1} = \frac{2r_i T_i + r_i m^2 \Delta r^2 (T_i - T_\infty) - T_{i-1} \left(r_i - \frac{\Delta r}{2} \right)}{r_i + \frac{\Delta r}{2}}$$

$$\therefore T_{i-1} = \frac{2r_i T_i + r_i m^2 \Delta r^2 (T_i - T_\infty) - T_{i+1} (r_i + \frac{\Delta r}{2})}{r_i - \frac{\Delta r}{2}}$$

c) since $-k \frac{dT}{dr} \Big|_{r=r_2} = h (T(r=r_2) - T_\infty)$

we can discretize $\frac{dT}{dr}$ as $\frac{\Delta T}{\Delta r} = \frac{T_i - T_{i-1}}{\Delta r}$

$h \rightarrow$ heat transfer coefficient

$k \rightarrow$ thermal conductivity of material

$T_\infty \rightarrow$ Boundary Temp

$T_\infty \rightarrow$ Ambient Temp

$$\therefore \frac{-k (T_i - T_{i-1})}{\Delta r} = h (T(r_2) - T_\infty)$$

$\therefore T_i$ is $T(r=r_2)$ in this case $\Rightarrow \frac{-k (T(r_2) - T_{i-1})}{\Delta r} = h (T(r_2) - T_\infty)$

$$\Rightarrow -k T(r_2) + k T_{i-1} = \Delta r h T(r_2) - \Delta r h T_\infty$$

$$\Rightarrow T(r_2) (\Delta r h + k) = k T_{i-1} + \Delta r h T_\infty$$

$$\therefore T(r_2) = \frac{T_{i-1} k + T_\infty \Delta r h}{(k + \Delta r h)}, \text{ or } \downarrow$$

$$\Rightarrow T_{i-1} = \frac{T(r_2) (k + \Delta r h) - T_\infty \Delta r h}{k}$$

boundary condition in recursion form

Solution Summary

a) Finite-Difference representation of ODE

$$r_i \left(\frac{T(r+\Delta r) + T(r-\Delta r) - 2T(r)}{\Delta r^2} \right) + \left(\frac{T(r+\Delta r) - T(r-\Delta r)}{2\Delta r} \right) - r m^2 \Delta r^2 (T(r) - T_\infty) = 0$$

b) Finite-Difference recursion form generalized

$$T_{i+1} \left(r_i + \frac{\Delta r}{2} \right) - T_i (2r_i + r_i m^2 \Delta r^2) + T_{i-1} \left(r_i - \frac{\Delta r}{2} \right) = T_\infty r_i m^2 \Delta r^2$$

Subsequent iteration

$$T_i = \frac{T_{i+1} \left(r_i + \frac{\Delta r}{2} \right) + T_{i-1} \left(r_i - \frac{\Delta r}{2} \right) + T_\infty r_i m^2 \Delta r^2}{(2r_i + r_i m^2 \Delta r^2)}$$

$$T_{i+1} = \frac{2r_i T_i + r_i m^2 \Delta r^2 (T_i - T_\infty) - T_{i-1} \left(r_i - \frac{\Delta r}{2} \right)}{r_i + \frac{\Delta r}{2}}$$

$$T_{i-1} = \frac{2r_i T_i + r_i m^2 \Delta r^2 (T_i - T_\infty) - T_{i+1} \left(r_i + \frac{\Delta r}{2} \right)}{r_i - \frac{\Delta r}{2}}$$

c) Boundary Condition recursion form

$$T_{i-1} = \frac{T(r_2) (K + \Delta r h) - T_\infty \Delta r h}{K}$$

If $T(r_2)$, T_∞ , h , K are given we can solve for T_{i-1} from boundary condition recursion form and then plug into recursion from B working back wards (i.e. $T(r_2) \Rightarrow T_{i+1} \xleftarrow{\text{from B.C. recursion}} T_{i-1} \xrightarrow{\text{from finite difference recursion}} T_i$) then subsequently solve for T_{i-1}

2) Governing Equation:

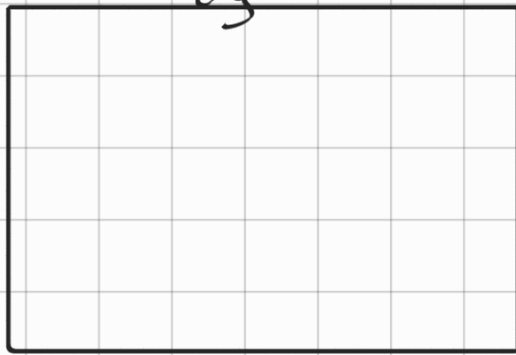
$$D \left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} \right) - Kc = 0 \quad \left| \begin{array}{l} D = 0.5 \\ K = 0.1 \end{array} \right.$$

$$\Rightarrow \left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} \right) - \frac{Kc}{D} = 0 \rightarrow C_{i,j}$$

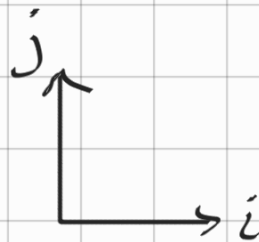
$$\frac{\partial c}{\partial y} = 0$$

$i = \text{index } x$

$j = \text{index } y$



$$\frac{\partial c}{\partial x} = 0$$



$$\frac{\partial c}{\partial y} = 0$$

from Finite Difference:

$$\frac{\partial^2 c}{\partial x^2} = \frac{C_{i-1,j} - 2C_{i,j} + C_{i+1,j}}{\Delta x^2}$$

$$\frac{\partial^2 c}{\partial y^2} = \frac{C_{i,j-1} - 2C_{i,j} + C_{i,j+1}}{\Delta y^2}$$

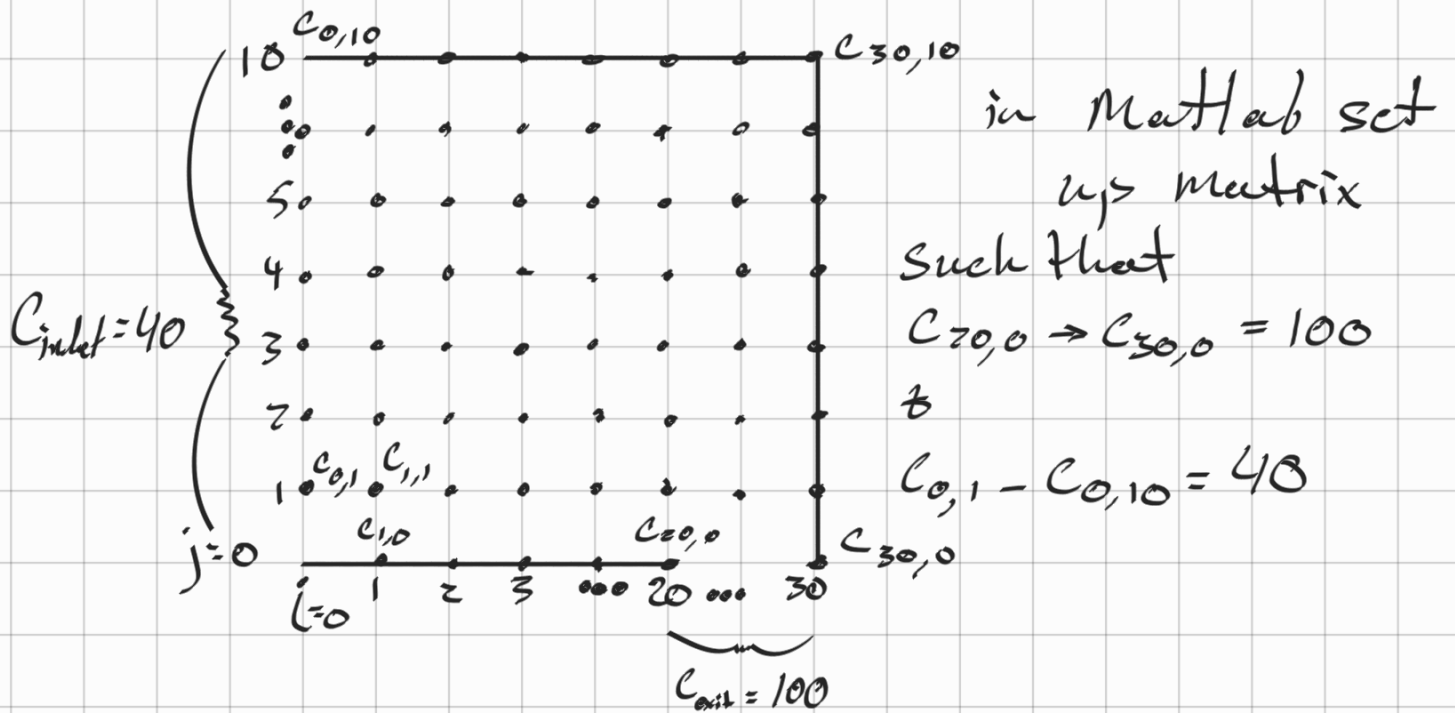
• Substituting in back into Governing Equation:

$$\frac{C_{i-1,j} - 2C_{i,j} + C_{i+1,j}}{\Delta x^2} + \frac{C_{i,j-1} - 2C_{i,j} + C_{i,j+1}}{\Delta y^2} - \frac{K C_{i,j}}{D} = 0$$

$$\text{use } \Delta x = \Delta y = 0.5 = h$$

$$\Rightarrow \frac{C_{i-1,j} - 2C_{i,j} + C_{i+1,j}}{h^2} + \frac{C_{i,j-1} - 2C_{i,j} + C_{i,j+1}}{h^2} - \frac{kC_{i,j}}{D} = 0$$

$$\Rightarrow C_{i+1,j} + C_{i,j+1} + C_{i-1,j} + C_{i,j-1} - \left(4 + \frac{k h^2}{D}\right) C_{i,j} = 0$$



```

clear
clc
clf
%%
%{
Consideering the Governing Heat Transfer Equation:


$$d^2C/dx^2 + d^2C/dy^2 - kc/D = 0$$


Using Finite Difference Method we can disretize to map

Using:
 $\Delta x = \Delta y = h = 0.5$ 

Assume:
    k=0.1
    D=0.5

%}
%% Setup
L = 30; %length of plate
W = 10; % width of plate
h = 0.5;
k = 0.1;
D = 0.5;

Ni = L/h+1; % number of elements in the i direction
Nj = W/h+1; % number of elements in the j direction

C_mat = zeros(Ni,Nj);
C_mat(1,:) = 40;
C_mat(40:end,1) = 100;

C = zeros(1,Nj-2);
%%
for t=1:100
    for i=2:Ni-1
        for j=2:Nj-1
            C_mat(i,j)=(C_mat(i+1,j)+C_mat(i,j+1)+C_mat(i-1,j)+C_mat(i,j-1))/(4+sqrt(
(k*h^2)/D);
        end
    end
end
C_mat(:,end)=C_mat(:,end-1);
C_mat(1:39,1)=C_mat(1:39,2);

```

```
C_mat(end,:)=C_mat(end-1,:);
```

```
end
```

```
C_mat_final=C_mat'
```

```
contourf(C_mat_final(1:end-1,1:end),25)
```

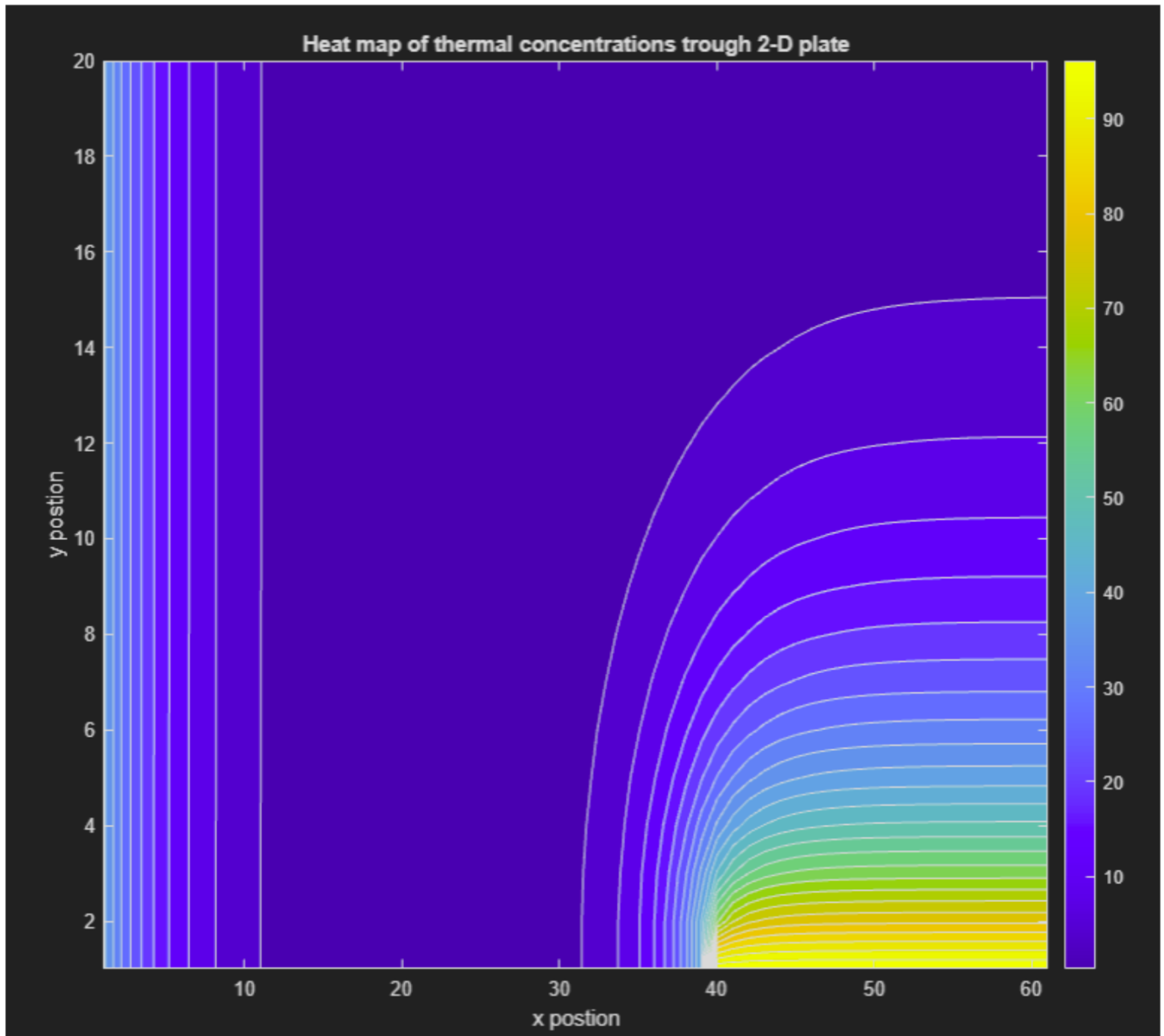
```
colorbar
```

```
title('Heat map of thermal concentrations trough 2-D plate')
```

```
ylabel('y postion')
```

```
xlabel('x postion')
```

```
%%
```



The advection-diffusion equation:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - u \frac{\partial c}{\partial x} - Kc$$

c = Concentration (mg/m^3)

t = time (min)

D = diffusion coefficient (m^2/min)

K = reaction rate (min^{-1}) where chemical decays

u = velocity (m/min)

Develop an Explicit scheme to solve (i.e. central Difference in space
- Forward Difference in time)

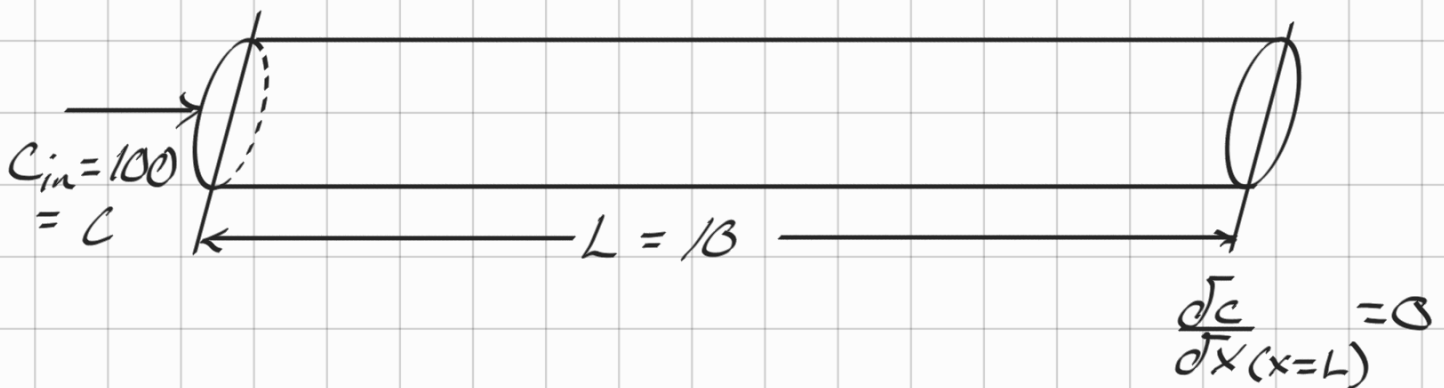
Test for $K=0.15$, $D=100$, $u=1$ Tank Length = 10 m

Using $\Delta x = 0.5$ $\Delta t = 0.001$ m

Assume: inflow concentration is 100

at the outlet concentration doesn't change w/ x -position

initial concentration of tank = 0



$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - u \frac{\partial c}{\partial x} - Kc$$

discretize w/r to central Difference in space & Forward Difference in time


```
clear
clf
clc
%%
%{
Consideering the Governing Advection-diffusion Equation:


$$dC/dt = Dd^2C/dx^2 - U dC/dx - kC$$


Develope an explicit scheme to solve:
    -Using-Central Difference in space
    -Forward difference in time

Varriables:
C = concentration (mg/m^3)
t = time (min)
D = diffusion (m^2/mm)
k = reaction rate (min^-1) whereas the chemical concentration decays
U = velocity (m/min)

Using:
     $\Delta x = 0.5 \text{ m} = h$ 
     $\Delta t = 0.001 \text{ min}$ 

Assume:
     $k = 0.15$ 
     $D = 100$ 
     $U = 1$ 
     $L = 10\text{m}$ 

%}
%% Setup
L=10;
h = 0.5;

dt = 0.001;
k = 0.15;
D = 100;
U = 1;

Ni = L/h+1; % number of elements in the i direction
Nt = (0:1:100); % number of time elements
xspan=(0:.1:10); % sets up disretiation of x to plot against distribution of✓
concentratrion
```

```

C_mat = zeros(length(Nt),Ni);

%setts intital condntions in matrix
C_mat(2,:) = 100;

for f=1:100 %iterates reccurrSION till values converge on final representation✓
of matrix
for t=2:length(Nt)-1 %iteration over time elements
    for i=2:Ni-1 %iteration over i distance elements

        %governing disrcetization formula
        C_mat(t+1,i)=C_mat(t,i-1)*dt*(D/h^(2)+U/2*h) +...
            C_mat(t,i)*(1-((2*D/h^(2))+k)*dt) +...
            C_mat(t,i+1)*dt*(D/h^(2)-U/2*h);
    % corrects for boundary condition
        C_mat(t,i-1)=C_mat(t,i);

    end
end
end

plot(xspan(1,1:end-2),C_mat(2:end-1,1))
title({'Advection-Diffusion Chemical Cononcentration Distribution after 100✓
mins'; ...
    'through 10 m Long rectangular reactor'})
ylabel('Chemcical Concentration (mg/m^3)')
xlabel('Postion in reactor (m)')

```

